# Transport Plan

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|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 512 megabytes |

There are $n$ cities and $m$ transport plans. For each integer $i$ in $[1, n-1]$, there is a bidirectional road between the $i$-th city and the $(i+1)$-th city with 1km length. The $i$-th transport plan is to transport $v_i$ ton goods from the $a_i$-th city to the $b_i$-th city.

It costs 1 yuan to transport 1 ton of goods for 1km. You can choose two cities $x, y$ and build a bidirectional portal between them. Goods can be transported between these two cities at no cost.

What is the minimum cost to accomplish all the $m$ transport plans?

## Input

The first line contains two integers $n, m$ ($2 \leq n \leq 3000, 1 \leq m \leq 2 \times 10^5$).

For the next $m$ lines, the $i$-th line contains three numbers $a_i, b_i, v_i$ ($1 \leq a_i, b_i \leq n, 1 \leq v_i \leq 10^6$).

## Output

One integer, denoting the minimum cost.

## Example

| standard input | standard output |
|---|---|
| 4 3<br>1 3 2<br>4 2 3<br>1 4 4 | 5 |